Plamen Markov



Senior Backend & AI Engineer — LLM Agents · MCP · Evals · REST APIs

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github.com/plamen911 · linkedin.com/in/plamen-markov-14254965 · Remote B2B · Part-time to full-time

PROFESSIONAL SUMMARY

Builds **LLM features into production software** and owns them after launch: a Claude plugin held to LLM-judge evals, a reporting agent that takes every figure from a named tool call, an MCP server published on npm, and vision OCR. Before that, **two decades of remote backend work**, most of it as the engineer responsible for the whole system, one platform for 13 years, from architecture and data model to rollout and maintenance. PHP and Laravel are the foundation; TypeScript and LLM integration are where the work is heading.

SELECTED MEASURABLE RESULTS

18 years — my first commercial product (2008), deployed at 10+ client firms, still running

20 years — continuous remote engagement with one US software studio (2005–2025)

86.5% — of a 60,000-line delivery platform authored personally; its lead for 13 years

4 systems — modernised from PHP 5.6 to modern Laravel / PHP 8.2

26,000+ — insurance policies migrated live, cutover parity-tested

436 tests — introduced to a 3-year-old production SaaS that had almost none

PHPStan 9 — plus an 878-test suite, gating every push in CI

AI ENGINEERING

AI-assisted development — how I write code

Claude Code every day for over a year inside **client production codebases**, not side projects: AI-assisted refactoring, test generation and debugging. The scarce skill is not generating code — it is reviewing what was generated, at senior depth, before it reaches paying customers.

AI inside the product — what I build

Designed and shipped a **tool-calling LLM reporting agent forbidden from computing any figure itself**: every number comes from one of fifteen curated tools with enum-constrained schemas, so the model picks from a fixed filter catalogue instead of free-forming a query. Gated behind an authorisation check and a feature flag because it surfaces personal data, and kept off the public MCP endpoint. Also vision-based document OCR (17 fields from a vehicle registration) and two MCP servers, one on npm automating the motor-insurance register check (case study: github.com/plamen911/go-insurance-checker; npm: go-insurance-checker). **The value here is constraint design, not ML research.** Separately I built and deployed a **RAG** product of my own end to end — **fire-expert.uk**, below.

WORK EXPERIENCE

Smartwary, Bulgaria — Senior PHP Developer (B2B contract) — **Aug 2025 – present**

Remote. Built **Browswave**'s pricing-intelligence platform until the client took it in-house in October 2026 — market intelligence for the consumer-electronics industry. 489 commits in 13 months; roughly 45% of tracked source.

Ivanov New Media Studio LLC — Nantucket, MA, USA — PHP Web Developer (contractor) — **2005 – 2025**

Remote throughout. Twenty years as the studio's long-term remote developer, building and maintaining production software for its US clients — among them Nantucket Energy, whose fuel-delivery platform I led for thirteen years.

Direct clients — Independent contractor — **2008 – present**

Remote. Sole developer on three production systems, all still in service and 100% my code: **Hipokrat** (hipokrat.net), an occupational-medicine application written in 2008 and since deployed for 10+ client firms — web-hosted, on-premises and standalone; **NPP Kozloduy**'s occupational-health system (since 2012); **General Broker**'s policy platform (17+ years). I gather the requirements directly from the business people who use them.

KEY PROJECTS

Browswave — PriceLens & PromoManager · multi-tenant pricing SaaS — **2025 – 2026**

Laravel 10 · PHP 8.2 · MySQL 8 · stancl/tenancy · Shopify REST + GraphQL · Docker · AWS

- **Database-per-tenant across 24 tenant databases** — consumer-electronics retailers and Shopify merchants across Bulgaria and the UK; the largest holds 35,500 products, 83,700 competitor records and roughly 18 million rows of pricing history.
- **Shopify rate-limit resilience across the tenant fleet** — proactive leaky-bucket pacing from the GraphQL cost extensions and the REST call-limit header, *Retry-After*-honouring retries with jittered backoff capped at 32 s, idempotent webhooks deduped on webhook ID, batched bulk mutations.

General Broker — insurance platform · sole developer, 100% of the code

2009 – present

Laravel 13 · PHP 8.4 · Livewire 3 · MariaDB · Redis · laravel/ai · Symfony 7.3 · React 18/19 · Bitbucket Pipelines · GitHub Actions

- **Made one place own the money.** Browser and server rounding disagreed by a cent, enough to block saves, so every premium, tax and total now comes from one server endpoint. Saves are transactional, non-destructive; invariant checks repair only under an explicit flag, after a snapshot.
- **Shipped a circuit breaker for the outbound integrations** after an upstream served an expired certificate for ten weeks. Three states in one cache entry, a lock so exactly one probe runs; wired to that upstream first, behind a seam the others join one at a time. 14 tests.
- **Built the property-quoting service alongside it** — a separate *Symfony 7.3* API (20 Doctrine entities, 57 routes, JWT with anonymous tokens), a React 18 quote wizard and a React 19 admin console.
- **Quality gates on both codebases** — PHPStan level 9, 878 tests and dependency audits gate every push on the Laravel platform; the Symfony service's first suite, 219 tests, surfaced an authorisation gap reaching admin data.

My own products — fire-expert.uk, Claude plugin, ADR Zone · sole developer

2026

Laravel 12 · PHP 8.5 · PostgreSQL 17 + pgvector · OpenAI + Cohere · Claude skills · n8n · React Native · Expo

- **Claude plugin drafting court-grade fire reports — simple cases 5–6 h to 10–15 min**, complex ones 7–8 h to 2–2.5 h with reviews, from phone photos; LLM-judge evals, PII checks in code.
- **Built and deployed a RAG product end to end, alone** — a Bulgarian-language retrieval-augmented generation assistant that gathers case facts conversationally, retrieves comparable past reports and emits a structured court-format expert report. Deployed on a 4 GB VPS: Docker Compose, TLS, nightly *pg_dump* backups, scripted *git archive* deploys with smoke tests.
- **ADR Zone — React Native / Expo app, live on the App Store** (apps.apple.com/bg/app/id6769773386). Offline dangerous-goods reference over a bundled 1,818-substance SQLite database, searched by name, UN number or placard code. Relevance ranking added after a substring match buried pure chlorine behind hundreds of chloride compounds - in an emergency-response tool that is a safety property, not a UX nicety - and a Cyrillic case-fold layer, because Hermes' *toLowerCase* and SQLite's *COLLATE NOCASE* are both ASCII-only.

Nantucket Energy — propane & heating-oil delivery platform · principal developer

2012 – 2025

PHP 8.1+ · Laravel 9 · Livewire · MySQL 8 (112 tables) · Alpine.js · Docker · Larastan · AWS S3

- **Thirteen years, two systems.** I wrote the original platform from scratch, then built its Laravel replacement and the bridge between them. **~52,000 of ~60,000 surviving lines (86.5%) and 1,481 of 2,499 commits** — principal author, active every year the project ran, verified by line-level *git blame*.
- **Integrated four independent tank-telemetry vendors across four transport styles** — hand-built SOAP envelopes to a .NET endpoint, two session-authenticated REST APIs, and a LoRaWAN webhook decoding raw hex payloads into battery voltage and tank percentage — plus the scheduled batch ingestion behind them.

NPP Kozloduy — occupational-health & medical-surveillance system · sole developer

2012 – present

PHP 8.2 · MySQL/MariaDB · Symfony components · Monolog · Pest · PHPStan · air-gapped intranet

- **Rebuilt in place from PHP 5.6 to PHP 8.2 with no rewrite and no flag-day cutover** — 114 domain services extracted from 172 flat page scripts, DTOs with Symfony Validator, Monolog and Pest. A parallel Laravel rewrite was started, evaluated and **rejected**, with the reasoning recorded in four ADRs including an explicit “no framework migration” decision.

TECHNICAL SKILLS

Backend	PHP 8.x (20+ yrs) · Laravel (expert) · **Symfony 7** (production API) · Doctrine ORM · Livewire · Node.js
Data	MySQL / MariaDB (expert) · **PostgreSQL 17 + pgvector** (vector similarity search) · query-plan & index diagnosis · Redis · SQLite · MongoDB (basic)
AI	Claude Code in production (daily, 1+ yr) · **RAG** (pgvector + full-text hybrid, Cohere rerank) · laravel/ai · MCP servers · tool-calling agent design · OpenAI vision OCR
Integrations	REST · GraphQL · SOAP · JSON-RPC 2.0 · webhooks · SFTP · SAML/OAuth flows · rate-limit resilience · Shopify · Amazon (Keepa)
Resilience	**circuit breaker (shipped)** · leaky-bucket rate-limit pacing · jittered backoff · idempotent webhooks · deadlock diagnosis & retry · queues · heartbeat alerting
Testing & quality	PHPUnit · Pest · Laravel Dusk · PHPStan / Larastan to level 9 · Pint · CI pipelines with dependency-audit gates
Mobile	**React Native 0.81** · Expo SDK 54 · TypeScript · offline SQLite · EAS build & submit (App Store)
Frontend & infra	Livewire · Alpine.js · **React 18/19** (two production front ends) · **Vue 2** · **Svelte 4** · Tailwind · Docker · AWS (EC2, S3) · nginx / php-fpm · Git

EDUCATION

SoftUni — PHP Web Developer, 2019 (894 h); Front-End Developer with JavaScript, 2019 (289 h)

TU Gabrovo — Computer Systems & Technologies, 2005

LANGUAGES & AVAILABILITY

Bulgarian — native · English — professional working proficiency

Always remote, B2B via a VAT-registered BG company; part-time to full-time.